Cuciureanu Dragos-Adrian
Grupa 231

# Homework 1

## Problem 1

We filter by DHCP. We see that the client requested IP 192.168.20.11 (No. 1254), but was Not Acknowledged (No. 1255). After that the client send a Discover (No. 1256) and received an Offer (No. 1257) for the IP 192.168.30.11, then the client Requested IP 192.168.30.11 (No. 1258), which was Acknowledged (No. 1259).

dhcp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1254 | 121.772905 | 0.0.0.0 | 255.255.255.255 | DHCP | 346 | DHCP Request - Transaction ID 0x5f511e61 |
| 1255 | 121.782156 | 192.168.30.1 | 255.255.255.255 | DHCP | 346 | DHCP NAK - Transaction ID 0x5f511e61 |
| 1256 | 121.805157 | 0.0.0.0 | 255.255.255.255 | DHCP | 346 | DHCP Discover - Transaction ID 0x96a1041e |
| 1257 | 121.810908 | 192.168.30.1 | 192.168.30.11 | DHCP | 361 | DHCP Offer - Transaction ID 0x96a1041e |
| 1258 | 121.821659 | 0.0.0.0 | 255.255.255.255 | DHCP | 346 | DHCP Request - Transaction ID 0x96a1041e |
| 1259 | 121.827910 | 192.168.30.1 | 192.168.30.11 | DHCP | 361 | DHCP ACK - Transaction ID 0x96a1041e |

The answer is: 192.168.30.11.

## Problem 2

We filter by NTP and see that the source (2003:41:6012:121::10) send a NTP packet (No. 2918) and then received one from the destination (2003:41:6012:110::dcf7:123), which had the mode flag set to Server.

ntp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 157 | 13.806387 | 192.168.121.40 | 212.224.120.164 | NTP | 94 | NTP Version 3, client |
| 158 | 13.808394 | 212.224.120.164 | 192.168.121.40 | NTP | 94 | NTP Version 3, server |
| 187 | 15.802944 | 192.168.121.40 | 78.46.107.140 | NTP | 94 | NTP Version 3, client |
| 188 | 15.809945 | 78.46.107.140 | 192.168.121.40 | NTP | 94 | NTP Version 3, server |
| 459 | 36.808209 | 192.168.121.40 | 148.251.154.36 | NTP | 94 | NTP Version 3, client |
| 460 | 36.814964 | 148.251.154.36 | 192.168.121.40 | NTP | 94 | NTP Version 3, server |
| 2918 | 286.367467 | 2003:51:6012:121::10 | 2003:51:6012:110::dcf7:123 | NTP | 114 | NTP Version 4, client |
| 2919 | 286.368969 | 2003:51:6012:110::dcf7:123 | 2003:51:6012:121::10 | NTP | 114 | NTP Version 4, server |
| 3891 | 334.812247 | 192.168.121.40 | 212.227.54.68 | NTP | 94 | NTP Version 3, client |
| 3893 | 334.817994 | 212.227.54.68 | 192.168.121.40 | NTP | 94 | NTP Version 3, server |

The answer is: 2003:41:6012:110::dcf7:123.

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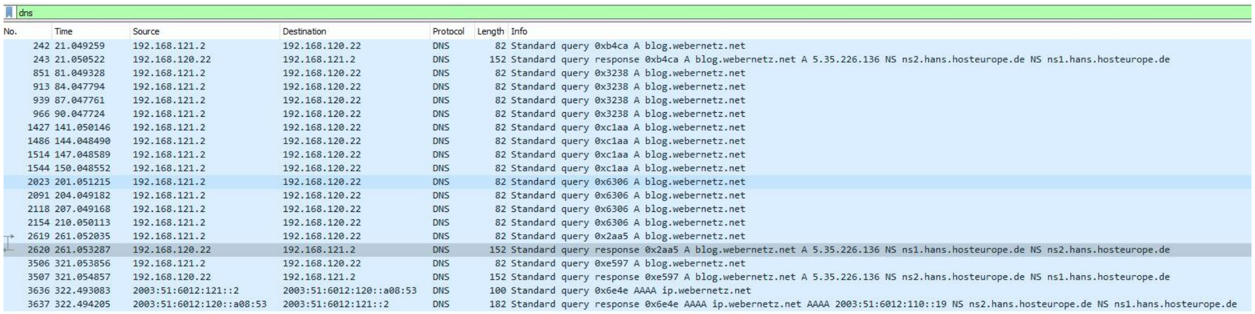
## Problem 3

We filter by DNS and see that in the pachet No. 2620, the authoritative name server for the domain blog.webergetz.net were ns1.hans.hosteurope.de and ns2.hans.hosteurope.de.



dns

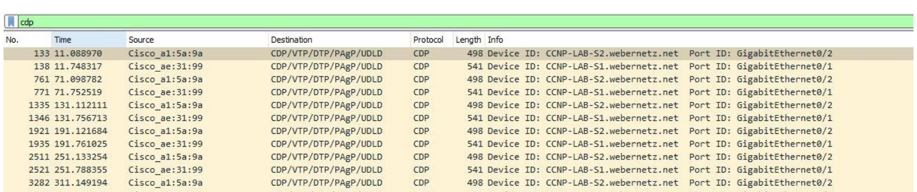
| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 242 | 21.049259 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0xb4ca A blog.webernetz.net |
| 243 | 21.050522 | 192.168.120.22 | 192.168.121.2 | DNS | 152 | Standard query response 0xb4ca A blog.webernetz.net A 5.35.226.136 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de |
| 851 | 81.049328 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x3238 A blog.webernetz.net |
| 913 | 84.047794 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x3238 A blog.webernetz.net |
| 939 | 87.047761 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x3238 A blog.webernetz.net |
| 966 | 90.047724 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x3238 A blog.webernetz.net |
| 1427 | 141.050146 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0xc1aa A blog.webernetz.net |
| 1486 | 144.048490 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0xc1aa A blog.webernetz.net |
| 1514 | 147.048589 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0xc1aa A blog.webernetz.net |
| 1544 | 150.048552 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0xc1aa A blog.webernetz.net |
| 2023 | 201.051215 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x6306 A blog.webernetz.net |
| 2091 | 204.049182 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x6306 A blog.webernetz.net |
| 2118 | 207.049168 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x6306 A blog.webernetz.net |
| 2154 | 210.050113 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x6306 A blog.webernetz.net |
| 2619 | 261.052035 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0x2aa5 A blog.webernetz.net |
| 2620 | 261.053287 | 192.168.120.22 | 192.168.121.2 | DNS | 152 | Standard query response 0x2aa5 A blog.webernetz.net A 5.35.226.136 NS ns1.hans.hosteurope.de NS ns2.hans.hosteurope.de |
| 3506 | 321.053856 | 192.168.121.2 | 192.168.120.22 | DNS | 82 | Standard query 0xe597 A blog.webernetz.net |
| 3507 | 321.054857 | 192.168.120.22 | 192.168.121.2 | DNS | 152 | Standard query response 0xe597 A blog.webernetz.net A 5.35.226.136 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de |
| 3636 | 322.493083 | 2003:51:6012:121::2 | 2003:51:6012:120::a08:53 | DNS | 100 | Standard query 0x6e4e AAAA ip.webernetz.net |
| 3637 | 322.494205 | 2003:51:6012:120::a08:53 | 2003:51:6012:121::2 | DNS | 182 | Standard query response 0x6e4e AAAA ip.webernetz.net AAAA 2003:51:6012:110::19 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de |

```
> Frame 2620: 152 bytes on wire (1216 bits), 152 bytes captured (1216 bits) on interface unknown, id 0
> Ethernet II, Src: Cisco_9e:11:41 (00:14:69:9e:11:41), Dst: Cisco_79:3f:11 (00:1e:7a:79:3f:11)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 121
> Internet Protocol Version 4, Src: 192.168.120.22, Dst: 192.168.121.2
> User Datagram Protocol, Src Port: 53, Dst Port: 56469
v Domain Name System (response)
    Transaction ID: 0x2aa5
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 2
    Additional RRs: 0
  > Queries
  > Answers
  v Authoritative nameservers
    > webernetz.net: type NS, class IN, ns ns1.hans.hosteurope.de
    > webernetz.net: type NS, class IN, ns ns2.hans.hosteurope.de
    [Request In: 2619]
    [Time: 0.001252000 seconds]
```

The answer is: ns1.hans.hosteurope.de.

## Problem 4

We filter by CDP and see in packet No. 133 (the first packet from CCNP-LAB-S2) that the Port ID for CCNP-LAB-S2 is GigabitEthernet0/2.



cdp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 133 | 11.088970 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 138 | 11.748317 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 761 | 71.098782 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 771 | 71.752519 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 1335 | 131.112111 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 1346 | 131.756713 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 1921 | 191.121684 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 1935 | 191.761025 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 2511 | 251.133254 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 2521 | 251.788355 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 3282 | 311.149194 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 3304 | 311.786290 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |

```
> Frame 133: 498 bytes on wire (3984 bits), 498 bytes captured (3984 bits) on interface unknown, id 0
> Ethernet II, Src: Cisco_a1:5a:9a (00:0a:8a:a1:5a:9a), Dst: CDP/VTP/DTP/PAgP/UDLD (01:00:0c:cc:cc:cc)
> 802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 1
> Logical-Link Control
v Cisco Discovery Protocol
    Version: 2
    TTL: 180 seconds
    Checksum: 0xde38 [correct]
    [Checksum Status: Good]
  > Device ID: CCNP-LAB-S2.webernetz.net
  > Addresses
  v Port ID: GigabitEthernet0/2
      Type: Port ID (0x0003)
      Length: 22
      Sent through Interface: GigabitEthernet0/2
```

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The answer is: GigabitEthernet0/2.

Problem 5

We filter by CDP and see in packet No. 133 (the first packet from CCNP-LAB-S2) that the IOS Software Version for CCNP-LAB-S2 is: Version 12.1(22)EA14.

cdp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 133 | 11.088970 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 138 | 11.748317 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 761 | 71.098782 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 771 | 71.752519 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 1335 | 131.112111 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 1346 | 131.756713 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 1921 | 191.121684 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 1935 | 191.761025 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 2511 | 251.133254 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 2521 | 251.788355 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |
| 3282 | 311.149194 | Cisco_a1:5a:9a | CDP/VTP/DTP/PAgP/UDLD | CDP | 498 | Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2 |
| 3304 | 311.786290 | Cisco_ae:31:99 | CDP/VTP/DTP/PAgP/UDLD | CDP | 541 | Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1 |

```
    Version: 2
    TTL: 180 seconds
    Checksum: 0xde38 [correct]
    [Checksum Status: Good]
  > Device ID: CCNP-LAB-S2.webernetz.net
  > Addresses
  > Port ID: GigabitEthernet0/2
  > Capabilities
  v Software Version
      Type: Software version (0x0005)
      Length: 276
      Software version: Cisco Internetwork Operating System Software
      Software version: IOS (tm) C2950 Software (C2950-I6K2L2Q4-M), Version 12.1(22)EA14, RELEASE SOFTWARE (fc1)
      Software version: Technical Support: http://www.cisco.com/techsupport
      Software version: Copyright (c) 1986-2010 by cisco Systems, Inc.
      Software version: Compiled Tue 26-Oct-10 10:35 by nburra
```

The answer is: Version 12.1(22)EA14.

Problem 6

I searched online “NVRAM config”and found out this:

- Network file systems (TFTP, rcp, and FTP)

So I wanted to search by TFTP and see which contain NVRAM, so I filtered by TFTP contains "NVRAM" and found this in packet No. 3770:

!

! Last configuration change at 20:55:45 UTC Fri Mar 3 2017 by weberjoh

! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by weberjoh

! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by weberjoh

version 15.1

service timestamps debug datetime msec

service timestamps log datetime msec

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service password-encryption

!

hostname CCNP-LAB-R2

!

boot-start-marker

boot-end-marker

!

!

enable secret 5 $1$Z.9j$Nvobsx9NvJzqtRLQqR.9b0

!

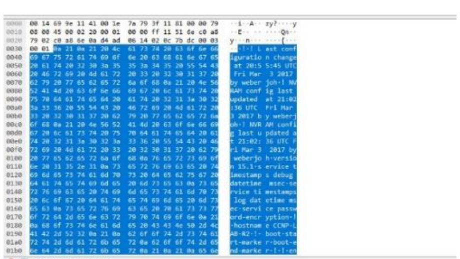
aaa new-model

!

!

aaa group server radius foobar

 server name blubb



The answer is: 21:02:36 UTC Fri Mar 3 2017.